Forecast-Informed Energy Storage Dispatch and Decision Recommendations for Renewable Integration in Ontario's Electricity Grid

As renewable generation becomes a larger share of Ontario's electricity system, grid operators face increasing difficulty in aligning variable wind and solar output with hourly demand. Because renewable generation is intermittent and uncertain, energy storage assets must be operated strategically to capture surplus renewable energy when it is available and release it when it is most useful to the system. This project investigates how short-term machine learning forecasts can support better operational storage decisions using publicly available electricity system data from the Independent Electricity System Operator (IESO). Using hourly wind generation, solar generation, demand, and related time-series variables, the project will develop an integrated decision-support framework that combines prediction, optimization, and operational recommendations. The objective is to determine whether forecast-informed storage dispatch can improve renewable utilization and reduce simulated system cost relative to simple rule-based policies.

The project will begin by preprocessing and structuring the IESO data as an hourly time-series dataset. Relevant predictive features will be constructed from historical renewable output, recent demand patterns, and temporal variables such as hour of day and day of week. Multiple forecasting models will then be developed to predict short-term wind and solar generation, with candidate approaches including baseline regression models and machine learning methods such as random forest or gradient boosting. Forecasting performance will be evaluated using held-out test periods and standard error metrics such as mean absolute error and root mean squared error.

These forecasts will then be integrated into a prescriptive optimization model for storage operation. The storage dispatch problem will be formulated as a linear programming or mixed-integer linear programming model that determines hourly charging, discharging, and state-of-charge decisions while satisfying operational constraints such as storage capacity, charge/discharge limits, and energy balance over time. If a mixed-integer formulation is used, binary decision variables may also be introduced to model mutually exclusive charging and discharging states. The optimization objective will be to maximize the effective use of renewable generation and minimize simulated system cost. The performance of the forecast-informed optimization model will be benchmarked against simpler alternatives such as no-storage operation and fixed rule-based dispatch strategies.

The final component of the project will translate model outputs into practical operational recommendations for a storage operator or grid decision maker. These recommendations will identify when storage should charge, discharge, or remain idle under different forecasted renewable conditions, and will highlight how improved forecasting quality changes prescriptive decisions. The anticipated result is that forecast-informed optimization will outperform naïve approaches by producing more effective storage schedules, increasing renewable utilization, and lowering simulated operating cost. More broadly, the project demonstrates how Management Engineering concepts can be integrated through a coherent workflow involving data preprocessing, machine learning, mathematical optimization, and decision recommendation to support more flexible and sustainable electricity system operations in Ontario.